

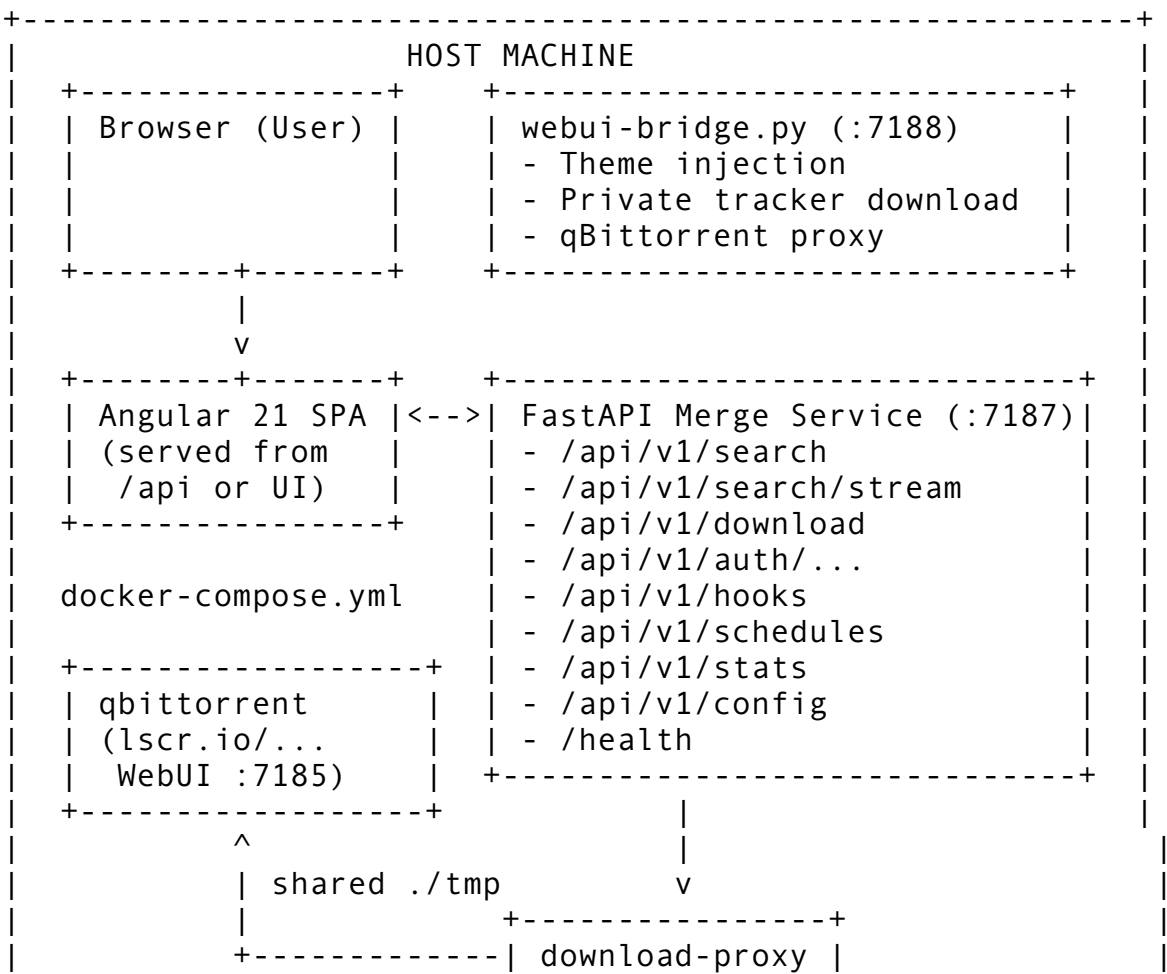
Dimension 01: Boba Project Architecture Deep Dive

Executive Summary

Boba-Base (GitHub: [milos85vasic/Boba-Base](#)) is a multi-tracker meta-search platform for qBittorrent, self-hosted and containerized. It consists of a FastAPI merge-search service, a Go/Gin backend (in parallel development), an Angular 21 SPA dashboard, a Python download-proxy, a private-tracker bridge (webui-bridge.py), Jackett integration, and a plugin system supporting 48+ torrent trackers. All runtime state is in-memory or JSON files – no relational database is used.

Key Finding for Browser Extension Integration: The FastAPI merge service on port 7187 exposes a full REST API (with OpenAPI/Swagger docs at /docs) including search, download, auth, hooks, scheduling, and SSE streaming. It serves the Angular SPA and supports CORS. The primary integration point for a browser extension is the POST /api/v1/search endpoint and GET /api/v1/search/stream/{search_id} SSE endpoint for real-time results, plus POST /api/v1/download to add torrents to qBittorrent.

1. Service Topology



```

|                                     | (Python 3.12, |                                     | |
| +-----+                         | also on 7186) |                                     |
| | jackson (:9117) | +-----+ |                                     |
| +-----+                         |                                     |
+-----+

```

Optional (docker-compose.quality.yml):

- SonarQube :9000 - Prometheus :9090 - Grafana :3000

New (Go backend, port 7187, profile: go):

- qbittorrent-proxy-go (qBitTorrent-go/)
- boba-jackson (:7189) - Jackson management API

Service Details

Service	Port	Container/Process	Language	Role
qBittorrent	7185	qbittorrent	C++ (upstream)	Core torrent client, WebUI
Merge Search	7187	qbittorrent-proxy	Python 3.12 (FastAPI)	Main API, search orchestration, Angular hosting
Download Proxy	7186	same container	Python	Legacy proxy, qBittorrent passthrough
WebUI Bridge	7188	Host process	Python 3	Private-tracker auth, theme injection
Jackson	9117	jackson	C#	External indexer aggregator
boba-jackson	7189	boba-jackson	Go (Gin)	Jackson management API
Go Proxy	7187	qbittorrent-proxy-go	Go (Gin)	Alternative merge search backend
Angular Dashboard	7187 (served)	static files	TypeScript/ Angular 21	SPA frontend

2. API Endpoints (Complete Catalog)

2.1 FastAPI Python Backend (Primary, Port 7187)

All endpoints served from the same FastAPI app in `download-proxy/src/api/__init__.py`.

Search Endpoints

Method	Path	Description	Request Body	Response
POST	/api/v1/search	Start async search (fire-and-forget)	SearchRequest	SearchResponse (status: running)
POST	/api/v1/search/sync	Blocking search (legacy)	SearchRequest	SearchResponse (full results)
GET	/api/v1/search/{search_id}	Get search status + results	-	SearchResponse
GET	/api/v1/search/stream/{search_id}	SSE stream of real-time results	-	text/event-stream
POST	/api/v1/search/{search_id}/abort	Cancel running search	-	{search_id, status}

Download Endpoints

Method	Path	Description	Request Body	Response
POST	/api/v1/download	Add torrent(s) to qBittorrent	DownloadRequest	{download_id, status, added_count, results}
POST	/api/v1/download/file	Download .torrent file directly	DownloadRequest	Blob (application/x-bittorrent)
POST	/api/v1/magnet	Generate magnet link from URLs	{result_id, download_urls}	{magnet, hashes}
GET	/api/v1/downloads/active	List active qBittorrent downloads	-	{downloads[], count}

Authentication Endpoints

Method	Path	Description	Request Body	Response
POST	/api/v1/auth/qbittorrent	Login to qBittorrent WebUI	{username, password, save?}	{status, version?, message?}
GET	/api/v1/auth/rutracker/status	Check RuTracker session status	-	{authenticated, status, message}
GET	/api/v1/auth/rutracker/captcha	Fetch CAPTCHA challenge	-	{captcha_required, captcha_image, captcha_token}

Method	Path	Description	Request Body	Response
POST	/api/v1/auth/rutracker/login	Solve CAPTCHA + login	CaptchaLoginRequest	{authenticated, message}
POST	/api/v1/auth/rutracker/cookie-login	Login with browser cookies	CookieLoginRequest	{authenticated, message}
GET	/api/v1/auth/status	All trackers auth status	-	{trackers: {rutracker, kinozal, nnmclub, iptorrents, qbittorrent}}
POST	/api/v1/auth/qbittorrent/logout	Clear saved qBittorrent creds	-	{status}

Hook Endpoints

Method	Path	Description	Request Body	Response
GET	/api/v1/hooks	List all hooks	-	{hooks[], count}
POST	/api/v1/hooks	Create a new hook	HookCreateRequest	HookResponse
DELETE	/api/v1/hooks/{hook_id}	Delete a hook	-	{message, hook_id}
GET	/api/v1/hooks/logs	Get hook execution logs	?limit=50&hook_name=	{logs[], count}

Scheduler Endpoints

Method	Path	Description	Request Body	Response
GET	/api/v1/schedules	List scheduled searches	-	{schedules[], count}
POST	/api/v1/schedules	Create scheduled search	ScheduleCreateRequest	Schedule object
GET	/api/v1/schedules/{id}	Get scheduled search	-	Schedule object
PATCH	/api/v1/schedules/{id}	Update scheduled search	ScheduleUpdateRequest	{id, name, enabled}
DELETE	/api/v1/schedules/{id}	Delete scheduled search	-	{deleted, schedule_id}

Theme Endpoints

Method	Path	Description	Request Body	Response
GET	/api/v1/theme	Get current theme	-	Theme state object
PUT	/api/v1/theme	Update theme	{paletteId, mode}	Updated theme state
GET	/api/v1/theme/stream	SSE theme updates	-	text/event-stream

Utility Endpoints

Method	Path	Description	Response
GET	/health	Merge service health	{status, service, version}
GET	/api/v1/bridge/health	WebUI bridge health probe	{healthy, status_code, bridge_url, port}
GET	/api/v1/config	Service URLs for dashboard	{qbittorrent_url, qbittorrent_internal_url, ...}
GET	/api/v1/stats	Search stats + tracker list	{active_searches, completed_searches, trackers[]}
GET	/docs	Swagger UI	Interactive API docs
GET	/openapi.json	OpenAPI 3.1 spec	Machine-readable schema

2.2 Go/Gin Backend (Alternative, Port 7187, Profile: go)

The Go backend (qBittorrent-go/cmd/qbittorrent-proxy/main.go) mirrors the Python API surface using the Gin framework. It is activated via docker-compose.yml profile go.

Key differences from Python backend: - Uses qBittorrent native search API (/api/v2/search/start) instead of direct plugin subprocess calls - SQLite database for system state (boba.db at /config/boba.db) - Admin/admin Basic Auth for mutating endpoints on boba-jackett (:7189)

Go boba-jackett API (port 7189):

Method	Path	Description
GET	/healthz	Health check
GET	/openapi.json	OpenAPI spec
GET/POST	/api/v1/jackett/credentials	Credential management
GET	/api/v1/jackett/indexers	List Jackett indexers

Method	Path	Description
GET	/api/v1/jackett/catalog	Catalog of available trackers
POST	/api/v1/jackett/catalog/refresh	Refresh catalog
GET	/api/v1/jackett/autoconfig/runs	List autoconfig runs
POST	/api/v1/jackett/autoconfig/run	Trigger autoconfig
GET/POST	/api/v1/jackett/overrides	Env variable overrides

2.3 qBittorrent WebUI API (Port 7185)

The native qBittorrent API is proxied through port 7186 (download-proxy) and 7188 (webui-bridge). Key endpoints used internally:

Method	Path	Description
POST	/api/v2/auth/login	Login (returns "Ok")
GET	/api/v2/app/version	Get qBittorrent version
POST	/api/v2/torrents/add	Add torrent (URLs or file upload)
GET	/api/v2/torrents/info	List all torrents
POST	/api/v2/search/start	Start qBittorrent search
GET	/api/v2/search/results	Get search results
POST	/api/v2/search/stop	Stop search
GET	/api/v2/search/plugins	List installed plugins

3. Request/Response Schemas

3.1 SearchRequest

```
{
  "query": "string (required, min_length=1)",
  "category": "string (default: 'all')",
  "limit": "integer (1-100, default: 50)",
  "enable_metadata": "boolean (default: true)",
  "validate_trackers": "boolean (default: true)",
  "sort_by": "string (default: 'seeds')",
  "sort_order": "string ('asc'|'desc', default: 'desc')"
}
```

3.2 SearchResponse

```
{
  "search_id": "uuid-string",
  "query": "original query",
  "status": "running|completed|failed|no_results|captcha_required",
  "results": ["SearchResultResponse"],
  "total_results": 0,
  "merged_results": 0,
  "trackers_searched": ["rutracker", "rutor", "1337x", ...],
  "errors": ["tracker: error message"],
  "tracker_stats": ["TrackerSearchStat"],
}
```

```
    "started_at": "ISO-8601 timestamp",
    "completed_at": "ISO-8601 timestamp|null"
}
```

3.3 SearchResultResponse

```
{
  "name": "torrent name",
  "size": "human readable or bytes",
  "seeds": 42,
  "leechers": 7,
  "download_urls": ["https://..."],
  "quality": "uhd_4k|full_hd|hd|sd|unknown",
  "content_type": "movie|tv|anime|music|game|software|ebook|other",
  "desc_link": "description URL",
  "tracker": "tracker name",
  "sources": [{"tracker": "...", "seeds": 42, "leechers": 7}],
  "metadata": {
    "source": "OMDb|TMDB",
    "title": "canonical title",
    "year": 2024,
    "poster_url": "https://...",
    "overview": "plot description",
    "genres": ["Action", "Sci-Fi"]
  },
  "freeleech": false
}
```

3.4 DownloadRequest

```
{
  "result_id": "merged result id (uuid)",
  "download_urls": ["https://tracker/download/...", "magnet:?xt=urn:btih:..."]
}
```

3.5 TrackerSearchStat

```
{
  "name": "tracker identifier",
  "tracker_url": "base URL",
  "status": "pending|running|success|empty|error|timeout|cancelled",
  "results_count": 42,
  "started_at": "ISO-8601|null",
  "completed_at": "ISO-8601|null",
  "duration_ms": 1234,
  "error": "error message|null",
  "error_type": "upstream_http_403|dns_failure|plugin_crashed|null",
  "authenticated": true,
  "attempt": 1,
  "http_status": 200,
  "category": "all",
  "query": "original query",
}
```

```
"notes": {}  
}
```

4. Authentication Mechanisms

4.1 qBittorrent WebUI Auth

- **Default credentials:** admin / admin (configurable via .env)
- **Login flow:** POST /api/v2/auth/login with form data username + password
- **Session:** Cookie-based (SID cookie returned)
- **Saved credentials:** Stored at /config/download-proxy/qbittorrent_creds.json (optional)
- **Endpoint:** POST /api/v1/auth/qbittorrent (proxy through merge service)

4.2 Private Tracker Auth

RuTracker: - Username/password from env (RUTRACKER_USERNAME, RUTRACKER_PASSWORD) - CAPTCHA challenge flow: GET /api/v1/auth/rutracker/captcha -> solve -> POST /api/v1/auth/rutracker/login - Cookie-based session stored in SearchOrchestrator._tracker_sessions - Alternative: Cookie login via POST /api/v1/auth/rutracker/cookie-login

Kinozal: - Username/password from env (KINOZAL_USERNAME, KINOZAL_PASSWORD) - Falls back to IPTorrents credentials if not set - Session-based authentication

NNM-Club: - Cookie-based only (NNMCLUB_COOKIES) - Very sensitive to proxies - Session stored in orchestrator

IPTorrents: - Username/password from env (IPTORRENTS_USERNAME, IPTORRENTS_PASSWORD) - Freeleech-only mode enforced - Session-based authentication

4.3 Jackett API Auth

- API key from env (JACKETT_API_KEY)
- Passed as query parameter or header to Jackett on port 9117

4.4 boba-jackett (Go) Auth

- **Admin Basic Auth:** admin / admin hardcoded
- GET/HEAD/OPTIONS pass through unauthenticated
- Mutating endpoints (POST/PATCH/DELETE) require Authorization: Basic YWRtaW46YWRtaW4=

4.5 CORS

- Configured via ALLOWED_ORIGINS env var (default: *)
 - All origins allowed in development
-

5. Data Models

5.1 Internal Dataclasses (Python, merge_service/search.py)

SearchMetadata â€œ The core search tracking object:

```
@dataclass
class SearchMetadata:
    search_id: str          # UUID
    query: str
    category: str = "all"
    started_at: datetime    # UTC
    completed_at: datetime | None
    total_results: int = 0
    merged_results: int = 0
    trackers_searched: list[str]
    errors: list[str]
    status: str = "running" # running|completed|failed|aborted
    tracker_stats: dict[str, TrackerSearchStat]
```

SearchResult â€œ Individual torrent result from a tracker:

```
@dataclass
class SearchResult:
    name: str
    link: str                # Download URL or magnet
    size: str                # Human-readable
    seeds: int
    leechers: int
    engine_url: str
    desc_link: str | None
    pub_date: str | None
    tracker: str | None
    category: str | None
    freeleech: bool = False
    content_type: str | None
    quality: str | None
```

MergedResult â€œ Deduplicated result combining multiple sources:

```
@dataclass
class MergedResult:
    canonical_identity: CanonicalIdentity
    original_results: list[SearchResult]
    total_seeds: int
    total_leechers: int
    best_quality: QualityTier | None
    download_urls: list[str]
    created_at: datetime
```

CanonicalIdentity â€œ Deduplication fingerprint:

```
@dataclass
class CanonicalIdentity:
```

```

infohash: str | None
title: str | None
year: int | None
content_type: ContentType | None    # Enum: movie, tv, anime, music, ...
season: int | None
episode: int | None
resolution: str | None
codec: str | None
group: str | None
metadata_source: str | None

```

ContentType Enum: movie, tv, anime, music, audiobook, game, software, ebook, other, unknown

QualityTier Enum: sd, hd, full_hd, uhd_4k, uhd_8k, unknown

5.2 JSON Persistence (No Relational DB)

File	Owner	Contents
/config/download-proxy/hooks.json	api/hooks.py	Array of HookConfig dicts
/config/download-proxy/qbittorrent_creds.json	api/routes.py	{username, password}
/config/merge-service/scheduling.json	merge_service/scheduler.py	Array of ScheduledSearch dicts
/config/merge-service/theme.json	api/theme_state.py	Theme palette + mode
/config/boba.db	Go backend only	SQLite (credentials, indexers, catalog, runs, overrides)

5.3 Go Models (qbittorrent-go/internal/models/)

Key Go structs mirror the Python dataclasses: - SearchRequest / SearchResponse / TrackerStat - DownloadRequest / DownloadResult / URLDownloadResult - QBittorrentAuthRequest / QBittorrentAuthResponse - Hook / Schedule / ThemeState

6. Plugin Architecture

6.1 Plugin System Overview

Boba uses the **qbittorrent nova3 plugin format** – Python classes implementing a standard interface. Plugins are installed to /config/qbittorrent/nova3/engines/.

Plugin execution flow: 1. SearchOrchestrator fans out to each enabled tracker 2. For public trackers: subprocess python3 -c <engine>.search() (isolated process) 3. For private trackers: Direct aiohttp calls with session cookies (Python) or via qbittorrent search API (Go) 4. Results parsed from stdout JSON (public) or HTML (private) 5. Deduplicator merges by infohash > title+size > fuzzy match

6.2 Plugin Interface

Each plugin implements:

```
class MyTracker:
    url = 'https://tracker.example.com'
    name = 'MyTracker'
    supported_categories = {'all': True, 'movies': '14', 'tv': '15', ...}

    def __init__(self):
        pass

    def search(self, what, cat='all'):
        """Return list of dicts: {link, name, size, seeds, leechers, engine}"""
        pass

    def download_torrent(self, url):
        """Download torrent file, return bytes or file path"""
        pass
```

6.3 Plugin Categories

48 built-in plugins organized into: - **Core plugins** (plugins/): rutracker, rutor, kinozal, nnmclub, iptorrents, 1337x, piratebay, yts, eztv, limetorrents, torlock, torrentgalaxy, solidtorrents, nyaa, kickass, megapeer, tokyotoshokan, bitsearch, gamestorrents, audiobookbay, rockbox, glotorrents, snowfl, bt4g, extratorrent, therarbg, etc. - **Community plugins** (plugins/community/): academic, ali213, anilibra, btsow, linuxtracker, pctorrent, pirateiro, torrentscsv, xfsb, yihua, yourbittorrent, etc. - **WebUI-compatible plugins** (plugins/webui_compatible/): kinozal, nnmclub, rutracker (specialized for WebUI bridge)

6.4 Public Tracker Registry (Python)

```
PUBLIC_TRACKERS = {
    "1337x": "https://1337x.to",
    "piratebay": "https://thepiratebay.org",
    "yts": "https://yts.lt",
    "rutor": "https://rutor.info/",
    "limetorrents": "https://www.limetorrents.lol",
    "torrentgalaxy": "https://torrentgalaxy.one",
    "nyaa": "https://nyaa.si",
    # ... 40+ more
}
```

Dead trackers (excluded by default, enable with ENABLE_DEAD_TRACKERS=1): ali213, audiobookbay, bitru, bt4g, btsow, extratorrent, eztv, one337x, pctorrent, solidtorrents, therarbg, torrentfunk, xfsb, yihua

6.5 Private Tracker Registry

```
PRIVATE_TRACKERS = {
    "rutracker": "https://rutracker.org",
    "kinozal": "https://kinozal.tv",
    "nnmclub": "https://nnm-club.me",
}
```

```
    "iptorrents": "https://iptorrents.com",  
}
```

7. Search Flow (Detailed)

7.1 Search Lifecycle

1. Client -> POST /api/v1/search {query, category}
2. FastAPI -> SearchOrchestrator.start_search()
3. SearchOrchestrator creates SearchMetadata with UUID
4. SearchOrchestrator._run_search() fires in background asyncio.Task
5. For each enabled tracker:
 - a. TrackerSearchStat created with status "pending"
 - b. Subprocess or aiohttp call initiated
 - c. Stat flips to "running"
 - d. Results collected into _tracker_results[search_id][tracker_name]
 - e. Stat flips to "success", "empty", "error", or "timeout"
6. Deduplicator.merge_results() combines by CanonicalIdentity
7. MetadataEnricher fills poster, overview, genres from OMDB/TMDB/AniList
8. Status flips to "completed"
9. Client polls GET /search/{id} or receives SSE events

7.2 SSE Event Types

Event	Payload	When
search_start	{search_id, status: "started"}	Stream begins
tracker_started	TrackerSearchStat	Tracker status pending -> running
tracker_completed	TrackerSearchStat	Tracker reaches terminal state
result_found	{search_id, name, seeds, leechers, tracker, size, link, content_type, quality}	New result discovered
results_update	{search_id, total_results, merged_results, trackers_searched}	Result count changes
search_complete	SearchMetadata.to_dict()	All trackers finished
error	{error, search_id}	Search not found or error
close	{search_id, reason}	Client disconnects

7.3 Deduplication Strategy

- Tier 1: Infohash match (exact, strongest)
- Tier 2: Normalized title + size match
- Tier 3: Levenshtein fuzzy title match
- Tier 4: Weak heuristic fallback

IPorrents freeleech results are **never merged** with non-freeleech sources (to preserve ratio).

8. Download Flow (Detailed)

8.1 Public/Magnet Downloads

1. Client -> POST /api/v1/download {result_id, download_urls}
2. API authenticates with qBittorrent via saved or env credentials
3. For each URL:
 - a. If tracker URL: fetch .torrent via orchestrator.fetch_torrent()
 - b. If magnet link: POST to /api/v2/torrents/add with urls param
 - c. If direct URL: POST to /api/v2/torrents/add with urls param
4. Return {download_id, status, added_count, results}

8.2 Private Tracker Downloads (webui-bridge.py)

1. Download request hits :7188 (webui-bridge)
 2. Bridge identifies tracker from URL patterns
 3. Bridge calls nova2dl.py <engine> <url>
 4. nova2dl imports plugin, calls download_torrent()
 5. Plugin authenticates with tracker (session/cookies)
 6. Plugin downloads .torrent to /shared-tmp/<uuid>.torrent
 7. Bridge uploads file to qBittorrent via multipart POST
 8. Bridge returns OK to client
-

9. Environment Variables (Complete Reference)

9.1 Required Variables

Variable	Description	Default
RUTRACKER_USERNAME	RuTracker login	(none)
RUTRACKER_PASSWORD	RuTracker password	(none)
KINOZAL_USERNAME	Kinozal login	(none)
KINOZAL_PASSWORD	Kinozal password	(none)
NNMCLUB_USERNAME	NNM-Club username	(none)
NNMCLUB_COOKIES	NNM-Club browser cookies	(none)
IPTORRENTS_USERNAME	IPorrents login	(none)
IPTORRENTS_PASSWORD	IPorrents password	(none)

9.2 Service Configuration

Variable	Description	Default
QBITTORRENT_HOST	qBittorrent hostname	localhost
QBITTORRENT_PORT	qBittorrent WebUI port	7185
QBITTORRENT_URL	Full qBittorrent URL	http:// localhost:7185

Variable	Description	Default
QBITTORRENT_USER / QBITTORRENT_USERNAME	WebUI username	admin
QBITTORRENT_PASS / QBITTORRENT_PASSWORD	WebUI password	admin
WEBUI_PORT	Internal WebUI port	7185
WEBUI_USERNAME	WebUI username (container)	admin
WEBUI_PASSWORD	WebUI password (container)	admin
PROXY_PORT	Download proxy port	7186
MERGE_SERVICE_PORT	FastAPI port	7187
BRIDGE_PORT	WebUI bridge port	7188
MERGE_SERVICE_HOST	FastAPI bind host	0.0.0.0
SERVER_PORT	Go server port	7187

9.3 Performance Tuning

Variable	Description	Default
MAX_CONCURRENT_SEARCHES	Max parallel searches	5
MAX_CONCURRENT_TRACKERS	Max parallel tracker calls	10
MAX_CONCURRENT_SSE_STREAMS	Max open SSE connections	32
PUBLIC_TRACKER_DEADLINE_SECONDS	Per-tracker timeout	15
PLUGIN_TIMEOUT	Plugin execution timeout	10
SSE_TIMEOUT	SSE connection timeout	30

9.4 Feature Toggles

Variable	Description	Default
ENABLE_DEAD_TRACKERS	Include dead trackers	0
DISABLE_THEME_INJECTION	Disable theme	0
JACKETT_API_KEY	Jackett API key	(none)
JACKETT_URL	Jackett URL	http://localhost:9117
ALLOWED_ORIGINS	CORS origins	*
LOG_LEVEL	Logging level	INFO

9.5 Metadata Enrichment

Variable	Description
OMDB_API_KEY	OMDb API key
TMDB_API_KEY	TMDB API key
ANILIST_CLIENT_ID	AniList client ID
TVDB_API_KEY	TVDB API key

9.6 RuTracker Specific

Variable	Description	Default
RUTRACKER_MIRRORS	Comma-separated mirror URLs	https:// rutracker.org,...
PENDING_CAPTCHAS_MAX	Max cached CAPTCHA challenges	1024
PENDING_CAPTCHAS_TTL_SECONDS	CAPTCHA challenge TTL	900

10. Frontend Architecture (Angular 21)

10.1 App Structure

```
frontend/src/app/  
  app.ts                # Root component (standalone)  
  app.config.ts         # App configuration  
  app.routes.ts        # Route definitions  
  components/  
    dashboard/          # Main search dashboard  
    qbit-login-dialog/  # qBittorrent login modal  
    confirm-dialog/     # Confirmation dialogs  
    magnet-dialog/     # Magnet link generator  
    theme-picker/       # Theme selection  
    toast-container/    # Toast notifications  
    tracker-stat-dialog/# Tracker status detail  
    site-footer/       # Footer  
  services/  
    api.service.ts      # HTTP client for all API calls  
    sse.service.ts      # Server-Sent Events handler  
    theme.service.ts    # Theme state management  
    dialog.service.ts   # Dialog management  
    toast.service.ts    # Toast notifications  
  models/  
    search.model.ts     # TypeScript interfaces  
    palette.model.ts    # Theme palette definitions  
  jaxett/  
    credentials/  
    indexers/  
    jaxett.routes.ts
```

10.2 API Service Pattern

The frontend uses Angular's `HttpClient` with a base URL of `' '` (same origin, since the SPA is served from the FastAPI app on `:7187`):

```
// api.service.ts  
private baseUrl = ''; // Same-origin requests  
  
search(req: SearchRequest): Observable<SearchResponse> {  
  return this.http.post<SearchResponse>(`${this.baseUrl}/api/v1/search`, r  
}
```

```
// SSE connection
connect(searchId: string): void {
    this.eventSource = new EventSource(`/api/v1/search/stream/${searchId}`);
    // Listens for: search_start, result_found, results_update, search_compl
}
```

10.3 Dashboard Features

- **Search form:** Query input, category dropdown, sort controls
 - **Live results table:** Real-time updates via SSE
 - **Tracker status chips:** Per-tracker progress indicators
 - **CAPTCHA modal:** RuTracker CAPTCHA challenge solving
 - **Download buttons:** Direct download, magnet link, file download
 - **Theme picker:** 8 palettes, dark/light mode
 - **qBittorrent login:** Modal for WebUI authentication
-

11. Hook System

11.1 Hook Events

Event	When Fired
search_start	Search initiated
search_progress	Intermediate results available
search_complete	All trackers finished
download_start	Download request received
download_progress	Download in progress
download_complete	All URLs processed
merge_complete	Deduplication finished
validation_complete	Tracker validation finished

11.2 Hook Registration

```
curl -X POST http://localhost:7187/api/v1/hooks \
-H "Content-Type: application/json" \
-d '{
    "name": "notify_on_download",
    "event": "download_complete",
    "script_path": "/config/scripts/notify.sh",
    "enabled": true,
    "timeout": 30,
    "environment": {"WEBHOOK_URL": "https://hooks.example.com/notify"}
}'
```

12. Browser Extension Integration Points

12.1 Primary Integration Strategy

A browser extension should communicate directly with the **FastAPI merge service on port 7187**. The extension can:

1. **Search:** POST /api/v1/search + GET /api/v1/search/stream/{id} (SSE)
2. **Download:** POST /api/v1/download to add to qBittorrent
3. **Get torrent file:** POST /api/v1/download/file
4. **Generate magnet:** POST /api/v1/magnet
5. **Check auth:** GET /api/v1/auth/status
6. **Get config:** GET /api/v1/config (for qBittorrent URL)

12.2 Extension API Usage Examples

```
// 1. Start a search
const searchResponse = await fetch('http://localhost:7187/api/v1/search',
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({ query: 'Ubuntu 24.04', category: 'all', limit: 50
});
const { search_id } = await searchResponse.json();

// 2. Stream results via SSE
const eventSource = new EventSource(
  `http://localhost:7187/api/v1/search/stream/${search_id}`
);
eventSource.addEventListener('result_found', (e) => {
  const result = JSON.parse(e.data);
  console.log(`Found: ${result.name} (${result.seeds} seeds)`);
});
eventSource.addEventListener('search_complete', () => {
  eventSource.close();
});

// 3. Download a torrent
await fetch('http://localhost:7187/api/v1/download', {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({
    result_id: 'uuid-of-result',
    download_urls: ['https://tracker.example.com/download/12345']
  })
});

// 4. Get torrent file as blob
const blobResponse = await fetch('http://localhost:7187/api/v1/download/file',
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({ result_id: 'uuid', download_urls: ['https://...']
});
const blob = await blobResponse.blob();
```

```
// 5. Check if service is running
try {
  const health = await fetch('http://localhost:7187/health');
  const { status } = await health.json();
  console.log('Boba is running:', status === 'healthy');
} catch (e) {
  console.log('Boba not available');
}
```

12.3 Content Script Integration

For injecting search/download buttons on torrent sites:

```
// manifest.json (v3)
{
  "host_permissions": ["*://1337x.to/*", "*://thepiratebay.org/*"],
  "permissions": ["storage", "activeTab"],
  "content_scripts": [{
    "matches": ["*://1337x.to/*"],
    "js": ["content.js"]
  }]
}

// content.js -- Send page torrents to Boba
const torrents = extractTorrentsFromPage();
chrome.runtime.sendMessage({
  action: 'search_result',
  torrents,
  source: window.location.hostname
});
```

12.4 Authentication Handling

The extension should: 1. Detect if Boba is running (GET /health) 2. Prompt for qBittorrent credentials if needed (POST /api/v1/auth/qbittorrent) 3. Handle RuTracker CAPTCHA challenges via GET /api/v1/auth/rutracker/captcha 4. Store credentials securely using browser extension storage API

12.5 CORS Considerations

- By default, ALLOWED_ORIGINS=* in development
- For production, set ALLOWED_ORIGINS=http://localhost:7187,chrome-extension://YOUR_EXTENSION_ID
- The FastAPI CORS middleware handles preflight OPTIONS automatically
- Extension background scripts are not subject to CORS restrictions

12.6 Recommended Extension Architecture

```
Browser Extension (MV3)
|
+-- background.js (service worker)
|   +-- Manages SSE connections to Boba
```

```

|   +-- Caches search results
|   +-- Handles downloads
|   +-- Stores credentials securely
|
+-- popup/ (React/Vue popup)
|   +-- Search form
|   +-- Results list
|   +-- Settings (Boba URL, credentials)
|
+-- content/ (content scripts)
|   +-- Injects Boba buttons on torrent sites
|   +-- Extracts torrent info from pages
|
+-- options/ (options page)
    +-- Full settings
    +-- Tracker configuration

```

13. Key Files Reference

File	Purpose
download-proxy/src/api/____init__.py	FastAPI app, lifespan, CORS, SPA serving
download-proxy/src/api/routes.py	Core search/download endpoints
download-proxy/src/api/auth.py	Tracker auth, CAPTCHA handling
download-proxy/src/api/streaming.py	SSE streaming implementation
download-proxy/src/api/hooks.py	Hook CRUD + dispatch
download-proxy/src/api/scheduler.py	Scheduled search management
download-proxy/src/merge_service/search.py	SearchOrchestrator, data models
download-proxy/src/merge_service/deduplicator.py	Result deduplication
download-proxy/src/merge_service/enricher.py	OMDB/TMDB metadata enrichment
webui-bridge.py	Private-tracker download bridge
qBitTorrent-go/cmd/qbittorrent-proxy/main.go	Go backend entry point
qBitTorrent-go/internal/api/search.go	Go search handlers
qBitTorrent-go/internal/service/merge_search.go	Go search service
frontend/src/app/services/api.service.ts	Angular HTTP client

File	Purpose
frontend/src/app/services/sse.service.ts	Angular SSE handler
frontend/src/app/models/search.model.ts	TypeScript type definitions
docker-compose.yml	Service topology
.env.example	Configuration template
docs/api/openapi.json	Machine-readable API spec

14. Version Numbers & Compatibility

Component	Version	Notes
Python	3.12-alpine	Download proxy container
FastAPI	Latest (pip)	Merge service framework
Angular	21	Frontend SPA
Go	1.21+	Backend (Gin framework)
qBittorrent	Latest (linuxserver image)	Core torrent client
Jackett	Latest (linuxserver image)	External indexer
SQLite	3.x	Go backend only

15. Security Considerations

1. **Hardcoded credentials:** Default admin/admin for qBittorrent WebUI and boba-jackett
 2. **Credential storage:** .env file is gitignored; container env vars used
 3. **CORS:** ALLOWED_ORIGINS=* in development; restrict in production
 4. **CAPTCHA TTL:** _pending_captchas uses TTLCache (15 min default)
 5. **Path traversal:** Hook script paths validated (no . . allowed)
 6. **Credential scrubbing:** Logging uses CredentialScrubber filter
 7. **No auth on read endpoints:** GET/HEAD pass through on boba-jackett
-

Citations

Claim: Boba-Base is a multi-tracker meta-search for qBittorrent with FastAPI
Source: Boba-Base README.md
URL: <https://github.com/milos85vasic/Boba-Base/blob/main/README.md>
Date: 2025-04-10
Excerpt: "Multi-tracker meta-search for qBittorrent -- self-hosted, contains
Context: Official project documentation
Confidence: high

Claim: The data model uses Python dataclasses and Pydantic models with no database
Source: Boba-Base docs/DATA_MODEL.md
URL: https://github.com/milos85vasic/Boba-Base/blob/main/docs/DATA_MODEL.md
Date: N/A (repo file)
Excerpt: "qBittorrent-Fixed has no relational database. All persistent state is
Context: Architecture documentation

Confidence: high

Claim: The SSE streaming endpoint emits result_found events as each tracker
Source: Boba-Base download-proxy/src/api/streaming.py
URL: <https://github.com/milos85vasic/Boba-Base/blob/main/download-proxy/src/api/streaming.py>
Date: N/A (repo file)
Excerpt: "yield SSEHandler.format_event(event='result_found', data={...})"
Context: Source code for SSE streaming
Confidence: high

Claim: The webui-bridge.py intercepts download requests for private trackers
Source: Boba-Base webui-bridge.py
URL: <https://github.com/milos85vasic/Boba-Base/blob/main/webui-bridge.py>
Date: N/A (repo file)
Excerpt: "This module solves the WebUI download issue by creating a bridge between the WebUI and the tracker"
Context: Private tracker bridge implementation
Confidence: high

Claim: The Go backend uses admin/admin Basic Auth for mutating endpoints
Source: Boba-Base qBitTorrent-go/internal/jackettapi/auth_middleware.go
URL: https://github.com/milos85vasic/Boba-Base/blob/main/qBitTorrent-go/internal/jackettapi/auth_middleware.go
Date: N/A (repo file)
Excerpt: "adminUser/adminPass mirror the project's hardcoded WebUI credentials"
Context: Authentication middleware source
Confidence: high

Claim: The download endpoint fetches torrents with tracker authentication
Source: Boba-Base download-proxy/src/api/routes.py
URL: <https://github.com/milos85vasic/Boba-Base/blob/main/download-proxy/src/api/routes.py>
Date: N/A (repo file)
Excerpt: "tracker = _is_tracker_url(url); if tracker: torrent_data = await tracker.fetch_torrent(url)"
Context: Download handler implementation
Confidence: high

Claim: There are 48 built-in search plugins supporting both public and private trackers
Source: Boba-Base docs/PLUGINS.md
URL: <https://github.com/milos85vasic/Boba-Base/blob/main/docs/PLUGINS.md>
Date: N/A (repo file)
Excerpt: "This project includes multiple search engine plugins for qBittorrent, uTorrent, and BitTorrent"
Context: Plugin documentation
Confidence: high

Claim: The OpenAPI spec is available at /openapi.json and frozen for CI deployment
Source: Boba-Base docs/api/openapi.json
URL: <https://github.com/milos85vasic/Boba-Base/blob/main/docs/api/openapi.json>
Date: N/A (repo file)
Excerpt: Full OpenAPI 3.1 spec with schemas for SearchRequest, SearchResponse, DownloadRequest, DownloadResponse, and UserRequest
Context: Machine-readable API specification
Confidence: high

Claim: The Angular frontend uses same-origin requests with empty baseUrl
Source: Boba-Base frontend/src/app/services/api.service.ts
URL: <https://github.com/milos85vasic/Boba-Base/blob/main/frontend/src/app/services/api.service.ts>
Date: N/A (repo file)

Excerpt: "private baseUrl = '';"
Context: Frontend API service
Confidence: high

Research completed. This document provides a comprehensive architectural analysis of Boba-Base suitable for browser extension development planning.